ENGINEERING CHANGE NOTE

ECN-001: Cyclic Executive Task Manager

ECN number	ECN-001
Document reference	comeauDulcey.CMPE311.ECN-001
Date	September 29, 2026
Originator	Dulcey Comeau, Lead Engineer, UMBC
Implements	ECO 0.1, 1/1/2030, Requestor: Prof. Kidd [2]
Project	PROJECT-CE (Project #2), based on PROJECT-ASYNC (Project #1)
Change type	Software architecture. No hardware or requirements change.
Distribution	Prof Kidd, Instructor; MD Safwan Zaman, TA

DESCRIPTION OF CHANGE

The PROJECT-ASYNC software [1] is modified to use a polled, round-robin cyclic executive task manager based on a function pointer table. The three asynchronous activities of PROJECT-ASYNC (blinking LED1, blinking LED2, and the serial-monitor dialog) become separate run-to-completion tasks that are dispatched by the task manager. The System's external behavior is unchanged.

REASON FOR CHANGE

Per ECO 0.1 [2]: the change results in a more reliable design and will reduce labor and design costs for implementing future extensions. With the function pointer table, a new task is added by writing one function and one table entry, without modifying the dispatcher.

CUSTOMER REQUIREMENTS

The customer requirements, C1 through C6, are identical to that of PROJECT-ASYNC. See [1] REQUIREMENTS section, Part I, Table 1.

HIGH-LEVEL TECHNICAL REQUIREMENTS

The high-level technical requirements, HL.1 through HL.6, are identical to that of PROJECT-ASYNC. See [1] REQUIREMENTS section, Part II, Table 3.

EDUCATIONAL CONSTRAINTS

The educational constraints, EC.1 through EC.5, and the example dialog are identical to that of PROJECT-ASYNC. See [1] REQUIREMENTS section, Part II, Tables 4 and 5. New educational requirements ER.1.0 through ER.1.2 are added by [4]; see [3] Table 2.

ENGINEERING CHANGES

Table 1. Documents and items changed

Document / item	Revision	Change
System Design Document: comeauDulcey.CMPE311.project1.AsynchTasking [1]	A to B	Superseded by comeauDulcey.CMPE311.project2.CyclicExecutive [3]. Title, references, glossary, software design and testing sections updated; all other sections reference [1].
Source code: projectAsync.ino	Replaced	Replaced by projectCE.ino. loop() now calls only dispatchTasks(). serviceLed() and handleInput() become taskLed1(), taskLed2() and taskSerial(), registered in taskTable[] and called through function pointers.
Testing and validation requirements: [1] Table 7	Extended	T.0-T.10 retained unchanged (ER.1.1). T.11-T.15 added (ER.1.1.1): function pointer dispatch, round-robin order and run-to-completion, dispatcher cycle time no greater than 5 ms, adding a task without changing the dispatcher, and invalid-input handling. See [3] Table 5.
Circuit schematic: [1] Figure 1, Table 6	Unchanged	No hardware change.
Customer and HL-requirements: [1] Tables 1 and 3	Unchanged	No change, per ECO 0.1.

DETAILED SOFTWARE CHANGES

- Added a task control block (Task: function pointer, period, last-run time, enable flag) and a three-entry task table, taskTable[].
- Added dispatchTasks(), a polled round-robin dispatcher that calls each enabled, due task through its function pointer in fixed index order.
- LED timing moved from the per-LED struct into the task table; each LED task now only toggles its pin. setLedInterval() writes only the selected LED's table row.
- Timing changed from lastToggle = now to fixed-rate scheduling (lastRun += period), with resynchronization after an overrun, to prevent long-term drift.
- Minimum accepted interval raised from 1 ms to 2 ms so a half-period of 0 ms cannot occur.
- Added an optional CE_INSTRUMENT build flag that reports worst-case cycle and task times over the serial-monitor for T.13. It is off in the delivered build.

IMPACT

Area	Impact
Hardware	None.
User interface	None. The serial dialog of [1] Table 5 is unchanged. The interval error message text is shortened.
Performance	No change to LED timing. Flash use 4,020 bytes (12.3%), RAM use 256 bytes (12.5%) on the ATmega328P.

Area	Impact
Testing	Full regression of T.0-T.10 plus new tests T.11-T.15 required on the new build.
Other designs	None known. Future features may be added as new task table entries.

EFFECTIVITY

Effective with the release of comeauDulcey.CMPE311.project2.CyclicExecutive (Revision B) and projectCE.ino to the project repository. All builds of the System after this date use the cyclic executive.

REFERENCES

[1] comeauDulcey.CMPE311.project1.AsynchTasking, "CMPE311 Project 1: Asynchronous Tasking Lab Report," Revision A.

[2] ECO 0.1, 1/1/2030, Requestor: Prof. Kidd (PROJECT-CE handout [4], Figure 1).

[3] comeauDulcey.CMPE311.project2.CyclicExecutive, "CMPE311 Project 2: Cyclic Executive Lab Report," Revision B.

[4] CMPE311 PROJECT#2: PROJECT-CE lab handout (projectTaskMgmtCE.pdf).

Repository: *[insert your GitHub repository URL here]*

APPROVAL

Role	Name	Signature / Date
Originator	Dulcey Comeau	
Approved (Instructor)	Prof. Kidd	